Do Multimodal Reasoning Models Talk Themselves into Hallucination? Self-Induced Inertia in Multi-Object Recognition

Anonymous ACL submission

Abstract

When a vision-language model labels several objects in one image, its own earlier answers can pull later ones toward the same class. We ask whether RL-trained reasoning VLMs amplify or correct this self-induced inertia: we compare same-base model pairs with and without reasoning post-training on ROPE, measure which error types reasoning changes, and test whether GRPO with a per-object reward and an inertia penalty reduces inertia.

1 Problem

ROPE (Chen et al., 2024) asks a VLM to name five objects marked by numbered boxes in one image, choosing each label from a candidate list. On its Adversarial subset (four objects of class A followed by one of class B), LLaVA-7B recognizes the B object 23.4% of the time when probed alone, 13.2% when all five are probed together, and 0.0% under teacher forcing, where the four preceding answers are replaced by the ground truth (unseen split). Since that context is *correct*, the failure is not error propagation: correct prior answers alone push the model onto a textual shortcut. We call this *self-induced inertia*.

These results predate reasoning VLMs such as MiMo-VL-RL and Qwen3-VL-Thinking, which reason at length before answering. Reasoning could amplify inertia ("boxes 1–4 are chairs around the table, so box 5 is probably another chair") or correct it ("box 5 is larger and flat; it is the table"). Evidence points both ways. Longer reasoning lowers attention to the image (Liu et al., 2025; Tian et al., 2026), and reasoning models rarely recover once an error enters the chain (Zhu et al., 2026). Yet self-verification is a stated goal of reasoning training. Among the 78 papers citing ROPE (Semantic Scholar, Oct. 2026), we found none that evaluates reasoning models. We ask:

RQ1 On ROPE, do reasoning versions hallucinate more or less than non-reasoning versions of the same base model, by class composition?

RQ2 On Adversarial items, how often is object 5 labeled A, how much of this is caused by the preceding answers, and does longer reasoning raise or lower it?

RQ3 Which errors does reasoning change: invented classes, classes borrowed from other objects in the image, or inertia?

RQ4 (stretch) Does GRPO with a per-object reward and an inertia penalty reduce inertia without hurting general reasoning?

2 Related Work

Object hallucination. POPE (Li et al., 2023) probes whether single objects exist. ROPE (Chen et al., 2024) queries five objects with controlled class composition and found the textual shortcut above, on 2024 non-reasoning VLMs. DOBench (Wang et al., 2026) separates perception errors from text-prior errors, and MIOH (Min et al., 2026) studies objects across several images. We study several objects within one image, on reasoning models.

Hallucination in reasoning VLMs. RH-Bench (Liu et al., 2025) finds that longer reasoning lowers visual attention and amplifies hallucination; Tian et al. (2026) find that RL reasoning harms basic perception; MM-THEBench (Huang et al., 2026) locates hallucinations inside reasoning traces. These report aggregate effects; we isolate one mechanism with controlled compositions and per-error-type metrics.

Inertia and error propagation. Sánchez Villegas et al. (2026) show that reasoning VLMs follow misleading text cues placed beside the image. Zhu et al. (2026) inject errors into reasoning chains and find that models self-correct in under 10%

of cases, and MM-Snowball (Jiang et al., 2026) shows early errors compounding across dialogue turns. All three study planted or erroneous context. In ROPE's Adversarial items the pull comes from the model's own *correct* answers, so methods that detect and repair earlier errors do not apply.

RL against hallucination. GRPO (Shao et al., 2024) scores each sample relative to its group. DEEPO (Zhuang et al., 2026) notes that uniformly wrong groups give no gradient on hard queries and adds expert prefixes. We address the same sparsity for multi-object outputs with partial credit, and target inertia directly in the reward.

3 Approach

Model pairs. MiMo-VL-7B SFT vs. RL (Xiaomi LLM-Core Team, 2025): the RL model adds one mixed on-policy RL stage to the SFT model, so their gap mostly reflects RL. Qwen3-VL-8B Instruct vs. Thinking (Bai et al., 2025): same base, with and without long reasoning. Together the pairs separate "RL changed the weights" from "the model thinks longer." Optionally, one API model with adjustable reasoning effort.

Probes. (i) *Default*: all five objects in one turn. (ii) *Single-object*: one object per turn. (iii) *Teacher forcing*, following ROPE: the prompt states the true labels of objects 1–4 and asks only for object 5, so the model reasons with that context; the gap to (ii) measures inertia from correct context. (iv) *Counterfactual prefix* (new): objects 1–4 are given a wrong class C; if object 5 drifts to C, the model copies its context rather than the image. (v) *Budget sweep*: thinking capped at several token budgets.

Metrics. Per-object accuracy by subset and position, and three error types: *invented* (a class absent from the image), *borrowed* (the class of another object in the image), and *inertia* (an Adversarial object 5 labeled A). We also measure think–answer consistency: whether the class the reasoning settles on matches the final answer.

Fix (RQ4). We train a 3B open VLM with GRPO on ROPE-style data under three rewards: all-or-nothing (1 only if all five labels are correct), per-object (fraction correct), and per-object minus 0.5 when an Adversarial B object is labeled A. All-or-nothing gives no signal when no sample is fully correct and punishes one error as hard as three; per-object credit alone can still give an inertia-only error a positive advantage, which the penalty removes. Rewards come from annotations, with no human or LLM judge. We check for regressions on MathVista (Lu et al., 2024).

4 Data

ROPE (Chen et al., 2024) is our test set, built from MSCOCO panoptic (Lin et al., 2014) and ADE20K (Zhou et al., 2019) images. Its subsets are Homogeneous (AAAAA), Heterogeneous (ABCDE), Adversarial (AAAAB), and In-the-Wild (mixed); its Seen/Unseen splits follow the source train/val split (about 10k rows on HuggingFace). Scoring is automatic, and we build on the official ROPE evaluation code. For RQ4 we generate ROPE-style training items from COCO and ADE20K annotations, excluding every ROPE image. We estimate 60–100 single-GPU A40 hours in total.

5 Plan and Milestones

- **Oct 27–Nov 2:** pilot MiMo-VL SFT vs. RL on 100 images per subset; confirm output parsing, the HuggingFace-to-Seen/Unseen split mapping, and a measurable gap.
- **Nov 3–15:** evaluation harness; Default and Single-object runs for both pairs on all subsets (RQ1).
- **Nov 16–29:** forcing probes, budget sweep, and error taxonomy (RQ2, RQ3); GRPO data and first training runs (RQ4).
- **Nov 30–Dec 6:** analysis and slides for the Dec 7/9 presentations.
- **Dec 7–16:** final runs, error analysis, and report.

RQ1–RQ3 are the core and stand alone if RQ4 slips; if the pilot shows no gap between versions, the budget sweep and forcing probes become the main axis.

6 Team and Division of Work

Four members, each owning one angle; all share analysis and writing.

- **M1:** evaluation harness; paired-model runs (RQ1).
- **M2:** budget sweep; think–answer consistency (RQ2).
- **M3:** error taxonomy; object size, salience, and frequency (RQ3).
- **M4:** forcing probes (RQ2); GRPO data, rewards, runs (RQ4).

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